

WORKSHEET #2 FOR MIDTERM 1: RELEVANT ALGEBRAIC DERIVATIONS

ECON 101–INTRODUCTION TO MICROECONOMICS

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*The purpose of this worksheet is for you to derive again some of the algebra results that might be relevant for you to know for the exam; **everything that appears in this worksheet, except perhaps a few derivations, has appeared in the past in my videos/notes.** You can use this worksheet as a study guide to organize your review; use it to look at specific points in the videos or a specific page in the lecture notes. When I design a question for the quizzes/midterm with random components, I rely on these formulas to compute valid solutions; therefore, this worksheet is an excellent way to prepare for the exam, albeit perhaps a little abstract. If you do not want to handle going over all this algebra, at least take a specific example, i.e., some specific values for the parameters, and go over these calculations.*

Suppose that the demand and supply equations are

$$Q_D = a - bP, \quad (1)$$

$$Q_S = c + dP, \quad (2)$$

where the parameters a, b, c and d satisfy

$$a > 0, \quad b > 0, \quad c \text{ is any real number}, \quad d > 0, \quad a - c > 0, \quad ad + cd > 0. \quad (3)$$

1. Using equations (1) and (2), find the equation of the inverse demand and the inverse supply, specifying their slopes, and their vertical and horizontal intercepts; analyze how they relate to the original parameters a, b, c and d . Note that what we always graph are the inverse demand and inverse supply.
2. Find the equilibrium price P^e and equilibrium quantity Q^e . Can you see why the restrictions in (3) are needed?
3. Calculate the point price elasticity of supply and demand at any point (P, Q) . Analyze how these elasticities relate to the original parameters, and to the slope of the inverse demand and supply. Now try to calculate these point elasticities at the market equilibrium (P^e, Q^e) and try to simplify the expression as much as possible.
4. Given two points (P_1, Q_1) and (P_2, Q_2) , calculate the arc price elasticity of demand and supply. How these elasticities relate to the point elasticities in the previous item?
5. Find expressions for consumer and producer surplus at the market equilibrium (P^e, Q^e) . Compute Social Surplus (consumer plus producer surplus).
6. Suppose that a tax $T > 0$ is imposed. What is the largest tax that can be imposed, in the sense that we would obtain sensible solutions to the after-tax quantity and prices? *Hint: Look at the graphs of the inverse demand and supply. This particular calculation doesn't appear in the lecture notes, but I always think about this when restricting the parameters that appear in the quizzes.*
7. Compute the price paid by consumers after the tax is imposed.
8. Compute the price received by sellers after the tax is imposed.
9. Compute the after-tax equilibrium quantity, Q_T .
10. Compute the government revenue from the tax.
11. Compute the tax incidence for consumers and producers. How do these relate both to the parameters of the demand and supply and to the point elasticities?

12. Compute the deadweight loss generated by the tax.
13. Compute the Consumer and Producer Surplus after the tax. Compute Social Surplus after the tax.
14. Compute the change in consumer surplus and producer surplus and the change in Social Surplus (change between before and after the tax.)
15. Check that the following identity holds $\Delta SS = R + DWL$ where ΔSS is the change in social surplus, R are government revenues, and DWL is deadweight loss. *Hint: we did this in the lecture via a numerical example and I also asked you to check this for each case in assignment 5 part 2. However, in here I am asking if you can provide the fully algebraic derivation.*
16. Now suppose that a subsidy $S > 0$ is imposed. What is the largest subsidy that can be imposed, in the sense that we would obtain sensible solutions to the after-subsidy quantity and prices? *Hint: Once again, look at the graphs of the inverse demand and supply. This particular calculation doesn't appear in the lecture notes, but I always think about this when restricting the parameters that appear in the quizzes.*
17. Compute the price paid by consumers after the subsidy is imposed.
18. Compute the price received by sellers after the subsidy is imposed.
19. Compute the after-subsidy equilibrium quantity, Q_S .
20. Compute the government expenditure from the subsidy.
21. Compute the subsidy incidence for consumers and producers. How do these relate to both the parameters of the demand and supply and to the point elasticities?
22. Compute the deadweight loss generated by the subsidy.
23. Compute Consumer and Producer Surplus after the subsidy. Compute Social Surplus after the subsidy.
24. Compute the change in consumer surplus and producer surplus and the change in social surplus (change between after and before the subsidy.)
25. Check that the following identity holds $\Delta SS + DWL = E$, where E is government expenditures, ΔSS is the change in social surplus and DWL is deadweight loss. *Hint: we did this in the lecture via a numerical example and I also asked you to check this for each case in assignment 5 part 2. However, in here I am asking if you can provide the fully algebraic derivation.*

Now you are going to analyze some of the extreme cases.

26. Suppose that in the equations for the demand (1) and supply (2), the demand is *perfectly inelastic* with $b = 0$ and the supply is a regular one, with $d > 0$. Compute everything from 1–26 again for this case. Pay particular attention to the equilibrium quantities before and after tax/subsidy, the elasticities, the tax/subsidy incidence and the deadweight loss. Note that consumer surplus is undefined in this case, as I explained in the lecture about consumer and producer surplus.
27. Suppose that in the equations for the demand (1) and supply (2), the demand is a regular one, with $b > 0$, but the supply is *perfectly inelastic*, with $d = 0$. Again, compute everything from 1–26 for this case, paying particular attention to the equilibrium quantities before and after tax/subsidy, the elasticities, the tax/subsidy incidence and the deadweight loss. Note that producer surplus is undefined in this case, as I explained in the lecture about consumer and producer surplus.

Now we will analyze the case of perfectly elastic demand or perfectly elastic supply.

28. A perfectly elastic demand corresponds to the case in which any quantity is consumed at a specific price P^* . Think about a horizontal line at P^* when graphing this demand equation. Note that the efficient equilibrium price will be precisely P^* . When supply is regularly sloped but demand is perfectly elastic, the price buyers pay will be $P^d = P^*$. The price sellers will receive will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that suppliers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic. Note that in this case Consumer surplus is zero and the elasticity of demand is undefined (it is infinity because the demand is a horizontal line.)

29. A perfectly elastic supply corresponds to the case in which any quantity is supplied at a specific price P^* . Think about a horizontal line at P^* when graphing this supply equation. Once again, note that the efficient equilibrium price will be P^* . When demand is regularly sloped but supply is perfectly elastic, the price sellers receive will be $P^s = P^*$. The price buyers will pay will be different from P^* and depends on whether we have a tax or a subsidy. Repeat the calculations that you have done in the previous parts. In particular, note that buyers will pay the whole tax/receive the whole subsidy since they are the side that is more inelastic. Note that in this case producer surplus is zero and the elasticity of supply is undefined (it is infinity because the supply is a horizontal line.)